

Linux Lab 6

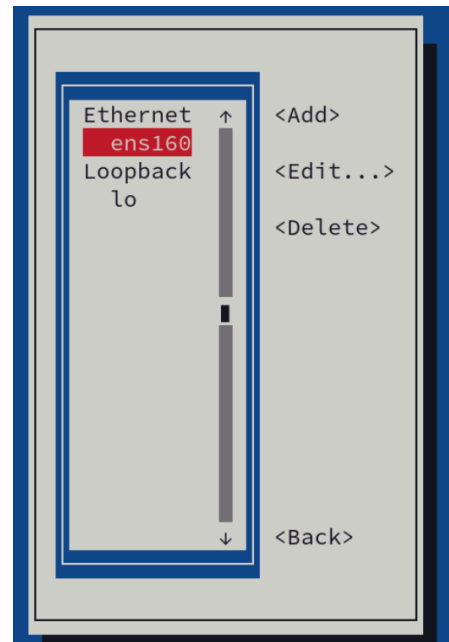
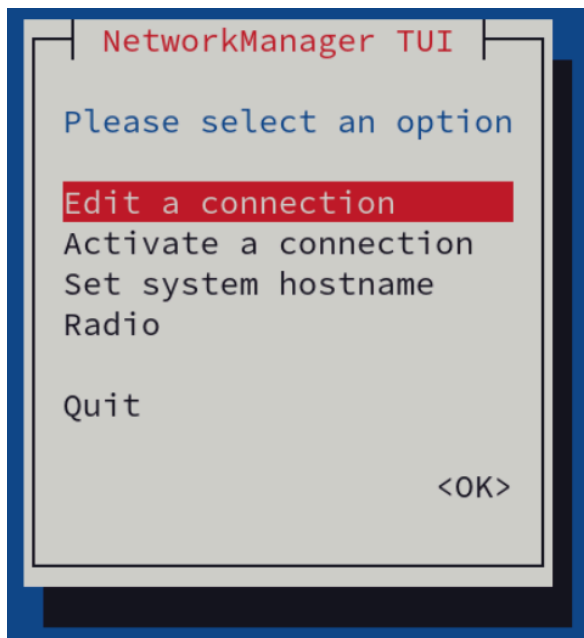
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Question 1: Setting up a DNS Server with an alias

1. Checking ip and gateway

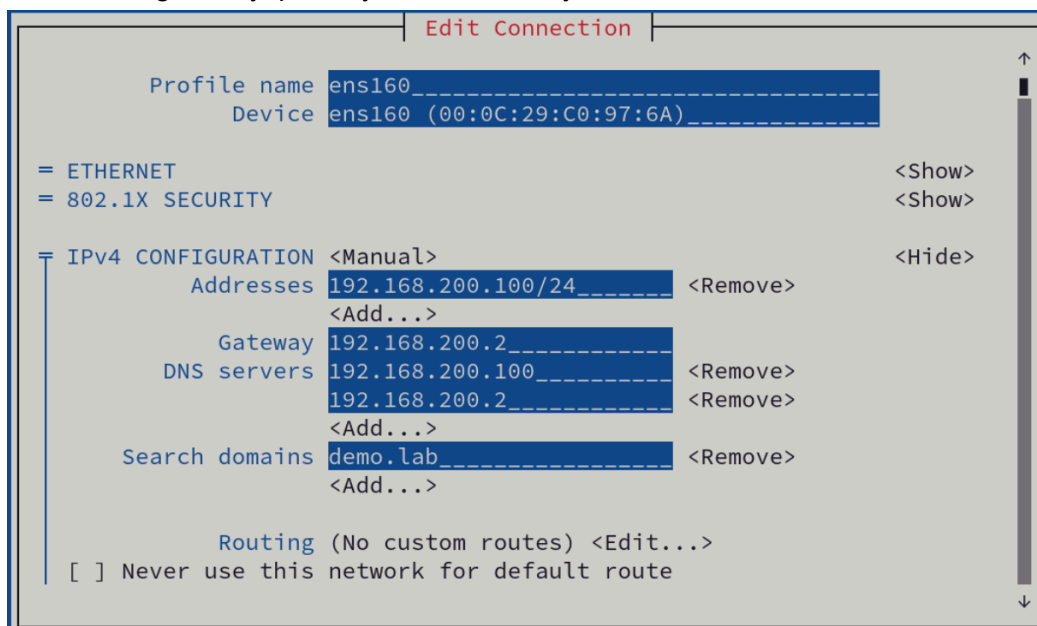
```
[csf@localhost ~]$ su
Password:
[root@localhost csf]# ip -br -c -4 a
lo                UNKNOWN    127.0.0.1/8
ens160            UP          192.168.200.132/24
[root@localhost csf]# route -n
Kernel IP routing table
Destination        Gateway            Genmask           Flags Metric Ref    Use Iface
0.0.0.0            192.168.200.2     0.0.0.0           UG    100    0      0 ens160
192.168.200.0      0.0.0.0           255.255.255.0     U     100    0      0 ens160
[root@localhost csf]#
```

2. Started network managed TUI using nmtui > Edit a connection

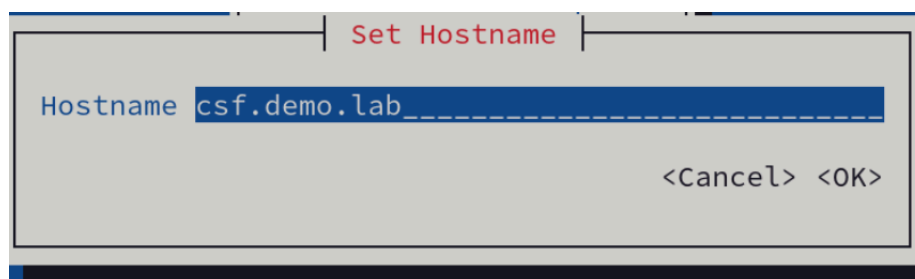


3. Set IPv4 Configuration from (Automatic to Manual)

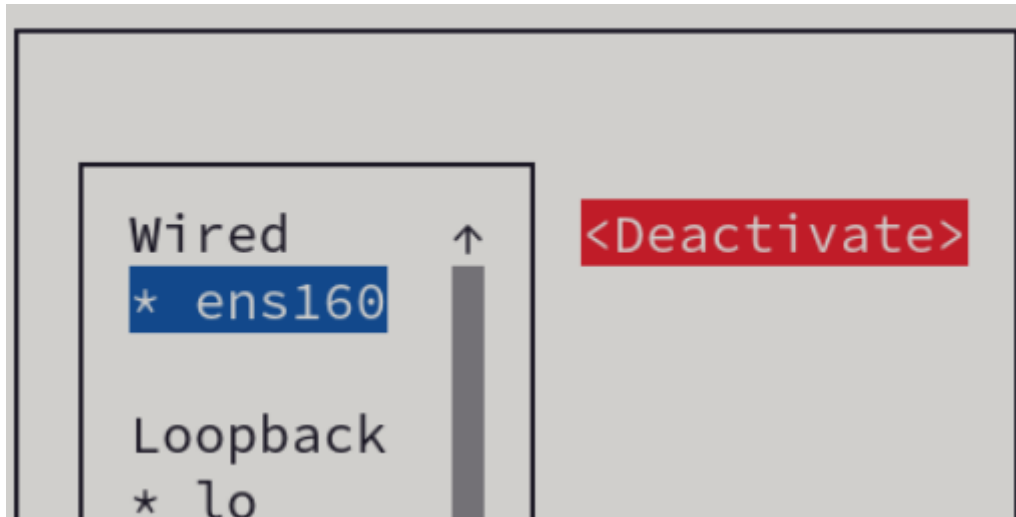
4. Added IP address, gateway, primary and secondary DNS, and search domains and saved



5. Updated the Hostname



6. Deactivated and re-activated the interface



7. Verified the changes

```
[root@localhost csf]# ip -br -c -4 a
lo                UNKNOWN          127.0.0.1/8
ens160            UP                192.168.200.100/24
[root@localhost csf]# route -n
Kernel IP routing table
Destination        Gateway            Genmask           Flags  Metric  Ref    Use  Iface
0.0.0.0            192.168.200.2     0.0.0.0           UG     100     0      0    ens160
192.168.200.0     0.0.0.0           255.255.255.0    U      100     0      0    ens160
[root@localhost csf]# hostname
csf.demo.lab
[root@localhost csf]# cat /etc/resolv.conf
# Generated by NetworkManager
search demo.lab
nameserver 192.168.200.100
nameserver 192.168.200.2
[root@localhost csf]#
```

8. Updated the name config file located at /etc/named.conf
9. Updated the name options - listen-on to DNS address and allowed-query to any

```
csf@localhost:/home/csf — vim /etc/named.conf

//
// Provided by Red Hat bind package to configure the ISC BIND named(8) DNS
// server as a caching only nameserver (as a localhost DNS resolver only).
//
// See /usr/share/doc/bind*/sample/ for example named configuration files.
//

options {
    listen-on port 53 { 192.168.200.100; };
    listen-on-v6 port 53 { ::1; };
    directory      "/var/named";
    dump-file       "/var/named/data/cache_dump.db";
    statistics-file "/var/named/data/named_stats.txt";
    memstatistics-file "/var/named/data/named_mem_stats.txt";
    secroots-file   "/var/named/data/named.secroots";
    recursing-file  "/var/named/data/named.recursing";
    allow-query     { any; };

    /*
     * - If you are building an AUTHORITATIVE DNS server, do NOT enable recursion.
     * - If you are building a RECURSIVE (caching) DNS server, you need to enable recursion.
     */
}
```

10. Added the forward and reverse query zones

```
csf@localhost:/home/csf — vim /etc/named.conf

file "named.ca";
};

include "/etc/named.rfc1912.zones";
include "/etc/named.root.key";
zone "demo.lab" IN {
    type      master;
    file      "for.demo.lab";
};

zone "200.168.192.in-addr.arpa" IN {
    type      master;
    file      "rev.demo.lab";
};
~
```

11. Restarted and verified the named service

```
[root@localhost named]# systemctl restart named
[root@localhost named]# systemctl status named
● named.service - Berkeley Internet Name Domain (DNS)
   Loaded: loaded (/usr/lib/systemd/system/named.service; disabled; preset: disabled)
   Active: active (running) since Thu 2026-09-03 21:22:05 IST; 6s ago
     Process: 5334 ExecStartPre=/bin/bash -c if [ ! "$DISABLE_ZONE_CHECKING" == "yes" ]; then /usr/sbin/named-c
     Process: 5336 ExecStart=/usr/sbin/named -u named -c ${NAMEDCONF} $OPTIONS (code=exited, status=0/SUCCESS)
    Main PID: 5337 (named)
      Tasks: 14 (limit: 48547)
     Memory: 38.9M (peak: 39.9M)
        CPU: 95ms
     CGroup: /system.slice/named.service
             └─5337 /usr/sbin/named -u named -c /etc/named.conf

Sep 03 21:22:05 csf.demo.lab named[5337]: network unreachable resolving './DNSKEY/IN': 2001:dc3::35#53
Sep 03 21:22:05 csf.demo.lab named[5337]: network unreachable resolving './NS/IN': 2001:dc3::35#53
Sep 03 21:22:05 csf.demo.lab named[5337]: network unreachable resolving './DNSKEY/IN': 2801:1b8:10::b#53
Sep 03 21:22:05 csf.demo.lab named[5337]: network unreachable resolving './NS/IN': 2801:1b8:10::b#53
Sep 03 21:22:05 csf.demo.lab named[5337]: network unreachable resolving './DNSKEY/IN': 2001:503:c27::2:30#53
Sep 03 21:22:05 csf.demo.lab named[5337]: network unreachable resolving './NS/IN': 2001:503:c27::2:30#53
Sep 03 21:22:05 csf.demo.lab named[5337]: network unreachable resolving './DNSKEY/IN': 2001:500:12::d0d#53
Sep 03 21:22:05 csf.demo.lab named[5337]: network unreachable resolving './NS/IN': 2001:500:12::d0d#53
Sep 03 21:22:05 csf.demo.lab named[5337]: network unreachable resolving './DNSKEY/IN': 2001:500:9f::42#53
Sep 03 21:22:05 csf.demo.lab named[5337]: network unreachable resolving './NS/IN': 2001:500:9f::42#53
lines 1-22/22 (END)
```

12. Added the forward file named as for.demo.lab

```
[root@localhost named]# vim for.demo.lab
[root@localhost named]# cat for.demo.lab
$TTL 1D
@      IN SOA csf.demo.lab. root@gmail.com. (
                                            0      ; serial
                                            1D     ; refresh
                                            1H     ; retry
                                            1W     ; expire
                                            3H )   ; minimum

@      NS      csf.demo.lab.
csf     A       192.168.200.100
portal  CNAME   192.168.200.100
[root@localhost named]#
```

13. Added the reversed file named as rev.demo.lab

```
[root@localhost named]# cp -arfv for.demo.lab rev.demo.lab
'for.demo.lab' -> 'rev.demo.lab'
[root@localhost named]# vim rev.demo.lab
[root@localhost named]# cat rev.demo.lab
$TTL 1D
@           IN SOA csf.demo.lab.      root@gmail.com. (
                                           0           ; serial
                                           1D          ; refresh
                                           1H          ; retry
                                           1W          ; expire
                                           3H )         ; minimum

           IN      NS      csf.demo.lab.
100        IN      PTR     csf.demo.lab.
[root@localhost named]#
```

14. Verified all the changes

```
[root@localhost named]# nslookup csf.demo.lab
Server:          192.168.200.100
Address:         192.168.200.100#53

Name:   csf.demo.lab
Address: 192.168.200.100

[root@localhost named]# nslookup portal.demo.lab
Server:          192.168.200.100
Address:         192.168.200.100#53

portal.demo.lab canonical name = csf.demo.lab.
Name:   csf.demo.lab
Address: 192.168.200.100

[root@localhost named]#
```

```
[root@localhost named]# host -t a csf.demo.lab
csf.demo.lab has address 192.168.200.100
[root@localhost named]# host -t a portal.demo.lab
portal.demo.lab is an alias for csf.demo.lab.
csf.demo.lab has address 192.168.200.100
[root@localhost named]#
```

```
[root@localhost named]# host -t soa 192.168.200.100
100.200.168.192.in-addr.arpa domain name pointer csf.demo.lab.
[root@localhost named]# host -t ns 192.168.200.100
100.200.168.192.in-addr.arpa domain name pointer csf.demo.lab.
[root@localhost named]# host -t ptr 192.168.200.100
100.200.168.192.in-addr.arpa domain name pointer csf.demo.lab.
[root@localhost named]#
```

```
[root@localhost named]# lsof -i:53
COMMAND  PID  USER   FD   TYPE DEVICE SIZE/OFF NODE NAME
named    5337 named   54u  IPv4  51585      0t0  UDP csf.demo.lab:domain
named    5337 named   55u  IPv4  51586      0t0  UDP csf.demo.lab:domain
named    5337 named   56u  IPv4  51587      0t0  UDP csf.demo.lab:domain
named    5337 named   57u  IPv4  51588      0t0  UDP csf.demo.lab:domain
named    5337 named   58u  IPv4  51589      0t0  TCP csf.demo.lab:domain (LISTEN)
named    5337 named   59u  IPv4  51590      0t0  TCP csf.demo.lab:domain (LISTEN)
named    5337 named   60u  IPv4  51591      0t0  TCP csf.demo.lab:domain (LISTEN)
named    5337 named   62u  IPv4  51592      0t0  TCP csf.demo.lab:domain (LISTEN)
named    5337 named   66u  IPv6  51593      0t0  UDP localhost:domain
named    5337 named   67u  IPv6  51594      0t0  UDP localhost:domain
named    5337 named   68u  IPv6  51595      0t0  UDP localhost:domain
named    5337 named   69u  IPv6  51596      0t0  UDP localhost:domain
named    5337 named   70u  IPv6  51597      0t0  TCP localhost:domain (LISTEN)
named    5337 named   71u  IPv6  51598      0t0  TCP localhost:domain (LISTEN)
named    5337 named   72u  IPv6  51599      0t0  TCP localhost:domain (LISTEN)
named    5337 named   73u  IPv6  51600      0t0  TCP localhost:domain (LISTEN)
[root@localhost named]#
```

Question 2

DNS (Domain Name System) is the phonebook of the internet, translating human-friendly domain names into computer-friendly IP addresses (like 142.250.190.46).

- Port Numbers: 53
- Protocols Used: UDP.

Question 3

Different records of DNS are:

- **A Record (Address):** Maps a domain name to an IPv4 address.
- **AAAA Record:** Maps a domain name to an IPv6 address.
- **CNAME (Canonical Name):** Creates an alias from one domain name to another.
- **MX Record (Mail Exchange):** Directs emails to a mail server.
- **NS Record (Name Server):** Specifies the authoritative name servers responsible for a domain.
- **PTR Record (Pointer):** Used for reverse DNS lookups, mapping an IP address back to a domain name.
- **SOA Record (Start of Authority):** Stores administrative and vital information about a DNS zone.
- **SRV Record (Service):** Defines a specific port and server for dedicated network services like VoIP or Active Directory.